Curling - Not For the Faint of Heart

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Abstract

In this paper we present two Partial Pooling Hamiltonian Monte-Carlo-Markov-Chain samplers designed to determine the relationship between Power-plays, ends, and play "success", in curling, with the odds of winning a curling match. Our model sought to prove some multiplicative relationship between any of these covariates and said odds of winning, and concluded favorably when it comes to the relationship between the specific end a power-play is made on, yet was unable to find any specific results when it comes to the relationship between "success" and winning.

Background and Objective

Throughout time immemorial, there have been activities that go against the grain of what are considered to be more traditional sports. Whether log rolling, where competitors attempt to not fall off of a spinning log, or even the curling of today, complete with brooms, ice, and stones, as long as you have someone to root for, anything can be a sport! Curling itself has it's own peculiar contexts making up the sport, thankfully not requiring quite so much balance! Whether it comes down to a team using a power-play to move some stones out of the way, to have a better chance at making the shot they need to win, or even the unique system put in place to evaluate how well a curler's shot matches their intended plan, there are many layers and wrinkles to the sport.

We have been charged by our professor to find some relationship between a couple of the more strategy related aspects of the sport. More specifically, we want to take a look at how exactly the odds of winning a match change depending on whether or not a team uses a power-play on a specific end, (Or play, in more general terms) and on top of that, whether or not these odds change based on whether the play made "failed in it's intended result."

A power play is a tactic used in mixed doubles curling where the pre-placed stones that would usually impede the movement of the teams' stones are moved out to one of the sides. A team is able to use a power play no more than once per match; its use is, however, not required. We aim to investigate how much of an impact on winning the power play has.

Another important aspect of our data that might not be clear with a surface level understanding of curling is the way that we measure "points" in our analysis. Rather than looking at points in the sense of how many a team may gain from landing their stone in the proper position, we are analyzing just how in line a given shot is with the intentions of the players, on a scale from 0 to 4, with 4 generally being a shot that is a complete success, 3 being close, and 0, 1, and 2 all being different levels of failure. We have further coerced this variable to represent this scale as a boolean for the success of a power-play.

Methodology

This analysis was completed using two Bayesian logistic regression models. Our outcome value across both was a binary indicator of whether the United States' mixed doubles curling team won their match. The likelihood level in both cases was taken to be a Bernoulli distribution that depends of the probability of success. This probability of success was estimated by a logistic regression model whose composition depended on the specifics of the question. Each of the β regression coefficients in the models were assumed to have a Normal distribution. These models were built with a random intercept but fixed effects; this allows there to be differences between matches while keeping the effects of the predictors consistent across matches. The use of fixed effects was necessitated by our sample size; there was not enough data to support the calculation of random slopes.

Our first research question explores the impact of the use of a power play on the odds success in a match. A formal model statement for this problem is as follows:

Let y_{ij} be a binary indicator of Team USA's success in mixed doubles curling in match $i = 1, ..., n_j$ in competition $j = 1, 2, 3, 4$. We model the probability of victory π_{ij} as via a logistic regression model that depends on: t_{ij}, which is an indicator for the presence of a power play on end five of a given match, v_{ij}, which is an indicator for the presence of a power play on end six of a given match, w_{ij}, which is an indicator for the presence of a power play on end seven of a given match, x_{ij} which is an indicator for the presence of a power play on end eight of a given match, and finally, z_{ij} which contains information regarding how many ends there were in a match. A formal statement of the model structure is as follows:

$$
\begin{aligned}
y_{ij} \mid \beta_{0j}, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5 &\sim Bernoulli\left(\pi_{ij}\right) \\
\log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) &= \beta_{0j} + \beta_1 t_{ij} + \beta_2 v_{ij} + \beta_3 w_{ij} + \beta_4 x_{ij} + \beta_5 z_{ij} \\
\beta_{0j} &\sim N\left(m_0, s_0^2\right) \\
\beta_1 &\sim N\left(-3.037, 0.1^2\right) \\
\beta_2 &\sim N\left(-1.35, 0.02^2\right) \\
\beta_3 &\sim N\left(-0.906, 0.012^2\right) \\
\beta_4 &\sim N\left(0.69, 0.025^2\right) \\
\beta_5 &\sim N\left(-2, 0.5^2\right) \\
m_0 &\sim N\left(2.19, 0.05^2\right) \\
1/s_0 &\sim Gamma(5, 100)
\end{aligned}
\tag{1}
$$

Our second research question also considers the success of both the power and non-power plays. A second formal model statement is as follows:

Once again, let y_{ij} be a binary indicator of Team USA's success in mixed doubles curling in match $i = 1, ..., n_j$ in competition $j = 1, 2, 3$. We model the probability of victory π_{ij} as a linear function of the presence of a power play within a match, x_{ij}, z_{ij}, w_{ij}, and t_{ij} all remain the same from the first model, while a_{ij} and b_{ij} represent a binary power-play success in a match, and the number of successful

non-power-plays in a given match, respectively.

$$
\begin{aligned}
y_{ij} \mid \beta_{0j}, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6 &\sim Bernoulli\left(\pi_{ij}\right) \\
\log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) &= \beta_{0j} + \beta_{1j} t_{ij} + \beta_{2j} v_{ij} + \beta_{3j} w_{ij} + \beta_{4j} x_{ij} + \beta_{5j} a_{ij} + \beta_{6j} b_{ij} \\
\beta_{0j} &\sim N\left(m_0, s_0^2\right) \\
\beta_{1j} &\sim N\left(-3.037, 0.1\right) \\
\beta_{2j} &\sim N\left(-1.35, 0.02\right) \\
\beta_{3j} &\sim N\left(-0.906, 0.012\right) \\
\beta_{4j} &\sim N\left(0.69, 0.025\right) \\
\beta_{5j} &\sim N\left(0.54, 0.25\right) \\
\beta_{6j} &\sim N\left(0.49, 0.25\right) \\
m_0 &\sim N\left(0.69, 0.025\right) \\
1/s_0 &\sim Gamma(25, 1000)
\end{aligned}
\tag{2}
$$

In order to progress with this analysis, some changes to the data were required. Initially, the data included one observation for each end (of which there are many in a match). The data was pivoted to have one observation per match with the information regarding ends within a match moved to individual columns.

For both research questions, our prior information was based on Team Canada's performance across the four competitions considered. Team Canada was chosen as for the prior as the US and Canada are neighboring countries who are often on a similar playing field in sports. The observed chance of victory for the relevant situations within each β was calculated and subsequently used to calculate the odds or odds ratio as necessary. This value was then converted to log odds for use as the prior mean for the appropriate β. The standard deviations were chosen somewhat arbitrarily with the goal of making the priors relatively strong due to the small sample size of our data. The priors for the Gamma distributed precision parameters were created based on an estimated ideal mean and mode.

This analysis was completed in RStudio (version 2026.01.0+392). Packages used include tidyverse (version 2.0.0) and rstan (version 2.32.7).

Analysis and Results

The Impact of a Power Play

The first step towards answering our research question is to ensure that our model is appropriate for use; in order to do this, we will evaluate our $\widehat{R}$ values, bulk effective sample size (ESS), and traceplots.

Table 1 below shows our $\widehat{R}$ and ESS values for all our values of β.

β	R-hat	ESS
$\beta_{0,1}$	1.000313	7709.254
$\beta_{0,2}$	1.000408	8026.177
$\beta_{0,3}$	1.000567	8068.817
$\beta_{0,4}$	1.000283	8264.817
β_1	0.9999677	14466.25
β_2	1.000143	13949.92
β_3	1.000068	14306.12
β_4	1.000421	15170.09
β_5	1.000725	5948.204

Table 1: Model 1 R-hat and ESS Values

There are absolutely no concerns with either of these metrics. The highest R-hat value (1.000725 for β_5) is well below the threshold. The ESS values are also far above the common minimum of 400.

Another aspect to consider is the traceplots. The traceplots for all four β_0 as well as β_2 are shown below. The unshown traceplots for β_1, β_3, β_4, and β_5 are extremely similar to those included and as such were included to limit unnecessary repetition.

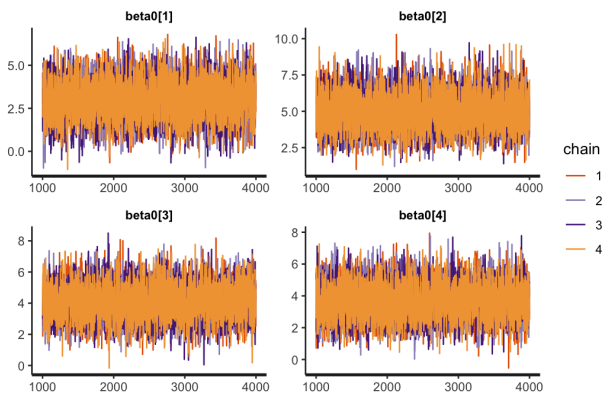

Figure 1: β_0 Traceplots

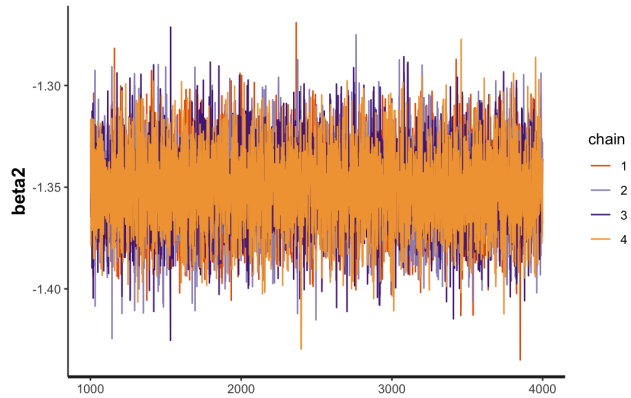

Figure 2: β_2 Traceplot

All the traceplots produced by this model show fast mixing. These traceplots, along with the R-hat and ESS values, confirm that this model has converged well enough to use it to answer our question.

Our β's of interest are β_1, β_2, β_3, and β_4 as these correspond to the presence of a power play on ends five through eight. The exponentiated versions of these β's are helpful in particular as they allow us to consider the odds scale. Table 2 below shows these exponentiated coefficients, as well as their confidence intervals.

e^{β}	Mean Estimate	95% Credible Interval
e^{β_1}	0.05	[0.04, 0.06]
e^{β_2}	0.26	[0.25, 0.27]
e^{β_3}	0.40	[0.39, 0.41]
e^{β_4}	1.99	[1.89, 2.09]

Table 2: Exponentiated Coefficients and Credible Intervals

"How do our intentions affect our winning?"

Now that we have taken a look at how our odds of winning change based on whether a team uses a power-play on a given end, we can take a look at how the "success" of the play dictates a teams odds of winning. Our trace-plot for random intercept as well as our two variables of interest from the second Bayesian logistic regression models follow as such:

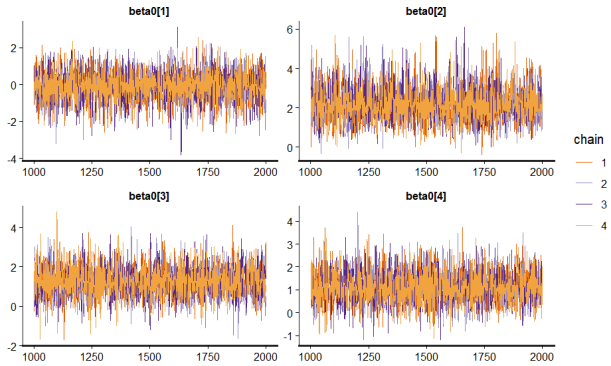

Figure 3: Trace-plots for β_{01}, β_{02}, β_{03}, and β_{04}

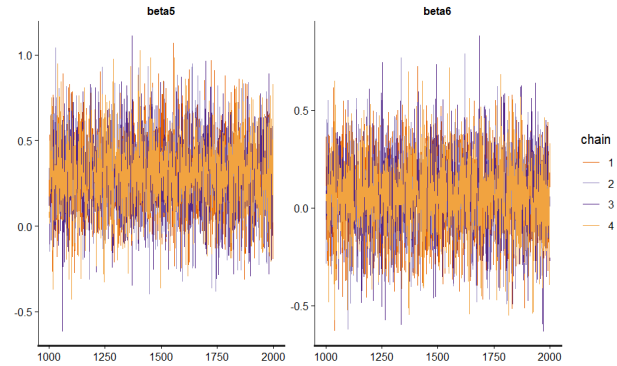

Figure 4: Trace-plots for β_5 and β_6

While these trace-plots are sufficient for our analysis, showing fast mixing, and having no issues with convergence, we shall still take a look at our $\hat{R}$ and bulk-ESS, to ensure that our model is sound.

β	R-hat	ESS
$\beta_{0,1}$	1.000008	4897.765
$\beta_{0,2}$	1.000008	4719.436
$\beta_{0,3}$	1.000008	4777.883
$\beta_{0,4}$	1.000441	4272.578
β_1	1.000846	5446.59
β_2	1.001924	5563.868
β_3	1.003489	5368.809
β_4	1.002306	5070.843
β_5	1.000559	4385.582
β_6	1.002016	4877.836

Table 3: Model 2 R-hat and ESS Values

As our values look good, seeing as we have no $\hat{R}$ above 1.1, and neither do we have any ESS below 400, we may be confident in the resulting coefficients from our model. As we have already seen the exponentiated β's 1-4 from our first model, we will skip the theatrics and solely exponentiate β_5 and β_6, given that the results of our first 4 β's are quite similar if not identical to those of model 1.

e^{β}	Mean Estimate	95% Credible Interval
e^{β_5}	1.349859	[0.860708, 2.138276]
e^{β_6}	1.0618365	[0.6976763, 1.5999942]

Table 4: Exponentiated Coefficients and Credible Intervals For Model 2

Discussion

The Impact of a Power Play

Considering power plays on an end by end basis allows us to answer a more nuanced question that considers the impact of the use of a power play on different ends. Interestingly, the impact of a power play being used on ends five through eight varies widely.

When considering matches with the same number of ends, the use of a power play on end five is associated with the odds of winning the match being 5% of the odds when a power play was not used on end five. Similarly, when considering matches with the same number of ends, the use of a power play on end six is associated with the odds of winning the match being 26% of the odds when a power play was not used on end six. End seven is unsurprisingly similar; when considering matches with the same number of ends, the use of a power play on end seven is associated with the odds of winning the match being 40% of the odds when a power play was not used on end seven.

The presence of a power play on these three ends seems to be associated with a decrease in the odds of winning. This is an interesting association, and one that is backed up by the 95% credible intervals of these coefficients. None of these e^{β}'s credible intervals include 1 - which would have indicated equal odds. Thus, we can be reasonably confident that the use of a power play on these ends is truly associated with a decrease in the odds of winning for Team USA.

Perhaps the most interesting end considered here is the eighth end; this end is actually associated with a multiplicative increase in odds of victory. When considering matches with the same number of ends, the use of a power play on end eight is associated with the odds of winning the match being 99% higher than the odds of winning when a power play was not used on end eight. The 95% credible interval for this exponentiated β also does not include 1 - though this time it lies entirely above the bound.

Taken all together, these results paint an interesting picture about the factors that go into the use of a power play in curling. It is important to consider that teams can only use a power play once per match. These results potentially suggest that Team USA uses a power play earlier in the match (ends five, six, and seven), when they are already behind. Thus, we hypothesize that they are potentially already in a position to lose and the chances that a power play could save them may be slim. On the other hand, when Team USA saves their power play for match eight - which is often the last match of the game - they are already in a position to win. Thus, the question of whether it's worth it to use a power play becomes a difficult one to answer. We argue that, although it appears to be associated with lower odds of winning, using a power play is likely worth it in ends five, six, and seven if it provides an outside shot at victory. There is less question about whether it's usage would be worth it in end eight as there's a clear associated increase in the odds of winning.

The Impact of Failure

When it comes to how the odds of winning a curling match change based on whether a play "failed in it's intended result," we can take a look at how our first model's results vary when also accounting for the "success" of a power-play, as well as the number of successful non-power-plays. Given the interpretation of a Bernoulli logistic regression being the multiplicative change in odds of one's outcome, while controlling for all other predictors, we find ourselves in a great spot to look at how the success of a power-play affects our earlier results.

Unfortunately, as our exponentiated credible intervals both contain one, we cannot necessarily so concretely discuss the effect of whether or not plays being successful affects our prior analysis. Given that we could have a reduction or increase in odds of winning, or even no change at all, it does not bode well to surmise.

While we may not be able to discuss our results as thoroughly on the numerical side of things, results such as ours justify a discussion on the validity of a measure of the player's intentions in the world of sports — Though that is not to say that it couldn't be that we simply don't have enough data to make inferences to that end — Measuring quantitatively just how much a player's attempt at a curling shot lines up with their intentions shot-wise winds up somewhat murky from the get-go, given that they themselves do not explicitly explain their intentions to whoever gives out these points, thus there exists some implied subjectivity in these measures. Along with this, we cannot account for the random nature of a player's

shot within this points measure, as no matter how good a player is, there will always be some effect upon their shot that is not accounted for, and thus a player could end up succeeding more than they intended too, or of course failing in the first place.

The Impact of Clustering

Our analysis considered matches grouped by competition, in an attempt to allow for some variation in the baseline odds of winning from competition to competition. This was backed up contextually as Team USA was made up of different people at different competitions. This grouping allowed for there to be some variation between the groups while still allowing for sharing of strength between them.

Due to sample size constraints, we were not able to include random effects on any β values, which did minimize the impact that the grouping had on the results as the effect of all our parameters was consistent across competitions.

With respect to our first research question, while there are relatively large differences in the exponentiated mean of each matches' respective β_0, their 95% credible intervals overlap each other frequently. Thus, it's entirely possible that the true order and values of the odds varies significantly from our results. There is a similar story with the exponentiated β values used to answer our second research question, most of our credible intervals did also overlap each other in our second model, showing that we cannot be too sure about the impact of clustering.

However, the conclusion should still be that the clustering had a positive impact on our analysis, as it allowed us to account for the variations between competitions, should they exist.

Limitations

The main limitation of our analysis is that we are working with a relatively small amount of data. In deciding to consider our data on the match level we reduced the dimensions of the data from 311 observations to just 41. This is a drastic reduction. In response to this, we were also forced to use relatively strong priors in order to obtain results.

The creation of these priors is also a potential limitation. They were created from data regarding Team Canada but this data was also subject to the small sample size issue. This required some averaging out to be done; for example, we did not consider the number of ends in a match while estimating priors for the presence of a power play on a specific end. Finally, the estimation of the prior standard deviations were slightly arbitrary; they were chosen to facilitate strong priors but their exact values were estimated.

Future Directions

The main future direction that this analysis could consider would be to add more data; we were working with only four curling competitions spread out over four years. There are more curling competitions that would provide further data for analysis - although the scale of the competitions may decrease once we are not considering the Olympics and World Championships.

We could also consider more countries. This analysis focused almost entirely on the United States with limited investigation into Canada for use as priors. Further investigations could consider multiple teams within a single competition to allow for further pooling of data.